TIKI-100 Emulator — User Manual

A friendly, step-by-step introduction to the TIKI-100 emulator. No prior experience with emulators or 1980s computers is required — every term and step is explained when it first appears.

If you are in a hurry, jump to **4. Quick start guide**.
Otherwise, read straight through.

> **Where this manual fits:** the project also has focused reference documents — INSTALL.md, USAGE.md, MENU.md, SERIAL.md, HARDWARE.md, CREDITS.md. This manual brings their content together as a beginner-friendly walkthrough.

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1. Introduction

What was the TIKI-100?

The **TIKI-100** was a Norwegian home and school computer launched in 1984 by **Tiki Data A/S** (Oslo). It was built around the **Zilog Z80A** processor running at 4 MHz — the same family of CPU that powered the Sinclair ZX Spectrum, Amstrad CPC, and many CP/M business machines of the same era.

What made the TIKI-100 stand out:

- **CP/M compatibility.** Its operating system, **TIKO/KP/M**, ran the same software as Digital Research's CP/M — meaning thousands of business and educational programs from across the world ran on it unchanged.
- **High-resolution graphics for its time** — up to 1024×256 pixels in two colours, plus medium and low-res colour modes.
- **Three-voice sound** via a General Instrument AY-3-8912 chip.
- **Adopted by Norwegian schools** after winning a major government tender in 1984, putting it on the desks of a generation of pupils.

It was originally called the *Kontiki 100* but renamed in 1984 after a naming dispute with explorer Thor Heyerdahl.

What does the emulator do?

An **emulator** is a program that pretends to be a different computer. The TIKI-100 emulator runs on a modern PC (Windows, Linux, macOS, or a Raspberry Pi) and behaves as if it were a real TIKI-100 from 1984. It runs the same ROM, the same operating system, and the same software, but inside a window on your modern desktop.

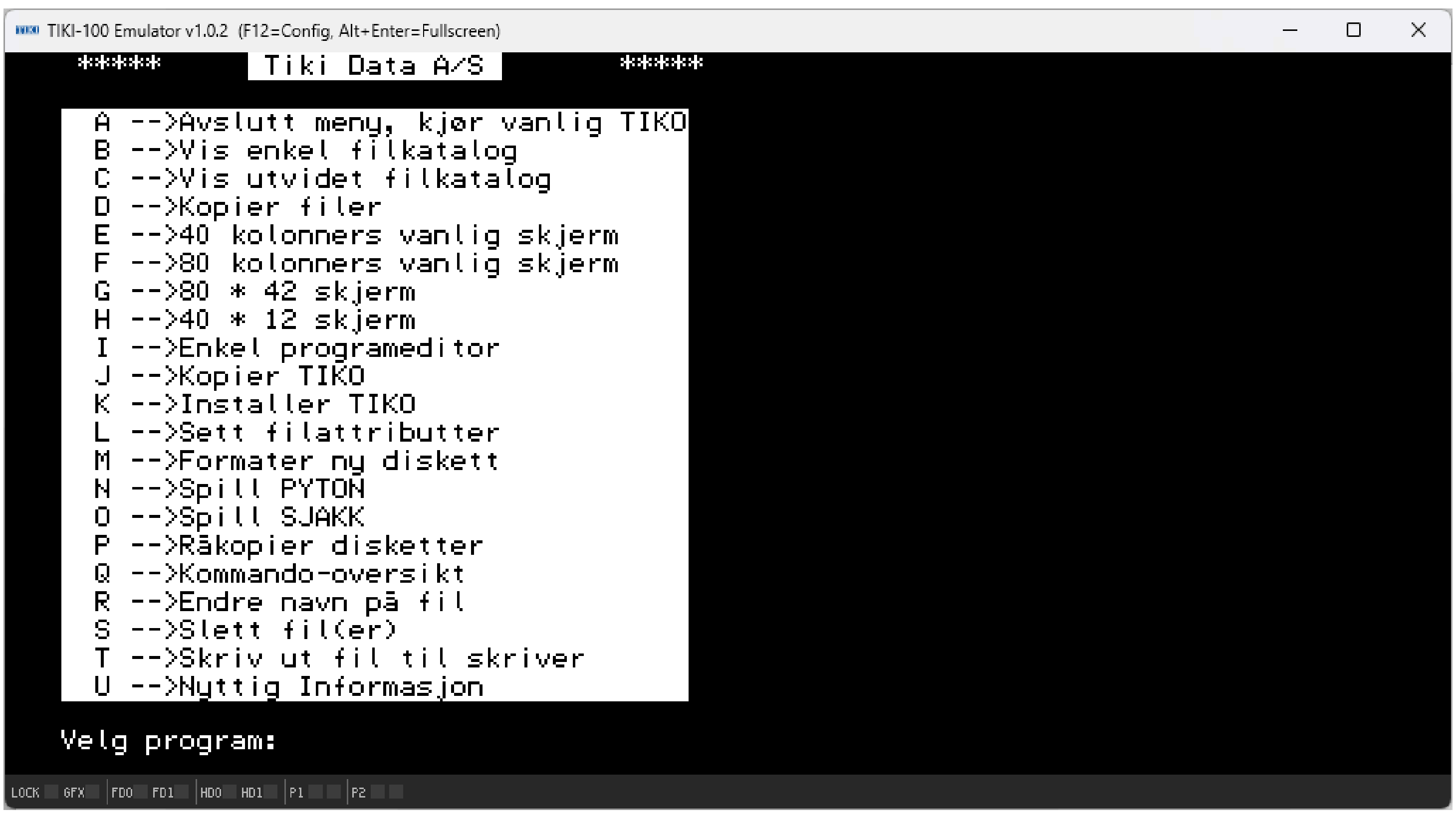
In practical terms, the emulator gives you:

- A full TIKI-100 system, including CPU, memory, video, sound, two floppy drives, two hard disks, two serial ports, and a printer port.
- The ability to load original disk images (`.dsk` files) and run authentic 1984 software.

- Modern conveniences like adjustable speed, save-and-load by file, an on-screen configuration menu, and TCP networking on the emulated serial ports (so you can dial real BBS-style services using the emulated Hayes modem).

Why use the emulator today?

- **Preservation.** Real TIKI-100 hardware is rare and ageing. The emulator keeps the platform alive.
- **Education.** Norwegian schoolchildren of the 1980s grew up with this machine. It is a piece of national computing history.
- **Software exploration.** Run TIKI-BAS, Tiki-Kalk, BRUM, WordStar, SuperCalc, dBase II, Turbo Pascal, and countless other CP/M classics.
- **Programming.** Z80 assembly, BASIC, Pascal, and C are all accessible through period-authentic toolchains.
- **Curiosity.** It is a small, self-contained 8-bit world you can explore from scratch in an afternoon.



2. Installing the emulator

This is a summary. The complete platform-by-platform reference is in INSTALL.md.

2.1 Windows

1. Visit the project's **Releases page** on GitHub.
2. Download `tiki100-windows-x64.zip` (modern 64-bit Windows) or `tiki100-windows-x86.zip` (32-bit).
3. Right-click the ZIP, choose **Extract All…**, and pick a folder (your desktop, `C:\Tools\tiki100\`, or a USB stick — anywhere works).
4. Open the extracted folder and double-click `tiki100.exe`.

> **Important — keep `SDL2.dll` next to `tiki100.exe`.** The release ZIP already has the right layout. If you copy the EXE somewhere else on its own, Windows will fail with *"SDL2.dll was not found"*. Copy the **whole folder**, not just the EXE.

2.2 Linux and Raspberry Pi

The easiest path on any Debian-based distribution (Debian, Ubuntu, Mint, Pop!_OS, **Raspberry Pi OS**) is the one-line installer:

```
curl -fsSL https://raw.githubusercontent.com/HackerCorpLabs/tiki100/main/scripts/install.sh | sh
```

After it finishes, run:

```
tiki100 -fd0 /usr/share/tiki100/disks/boot/tiko_kjerne_v4.01.dsk
```

Prefer not to pipe a shell script? Download the `.deb` for your architecture from the Releases page and install it manually — see INSTALL.md.

2.3 macOS (Apple Silicon)

1. Install SDL2 if you don't have it: `brew install sdl2`
2. Download `tiki100-macos-arm64.tar.gz` from the Releases page.
3. Extract and run:

```
tar xzf tiki100-macos-arm64.tar.gz
cd tiki100-macos-arm64
xattr -d com.apple.quarantine tiki100   # bypass Gatekeeper warning
./tiki100 -fd0 disks/boot/tiko_kjerne_v4.01.dsk
```

The `xattr` line is needed because the binary is not currently signed with an Apple Developer certificate.

2.4 What's included

When you install, you get:

File / folder	Purpose
`tiki100` (or `tiki100.exe`)	The emulator program itself
`SDL2.dll` (Windows only)	Graphics, sound, input library
`rom/`	The TIKI-100 system ROMs (V1.30, V1.35, V2.03 W)
`disks/boot/`	Bootable floppy images, including the standard `tiko_kjerne_v4.01.dsk` system disk

2.5 ROMs and legal considerations

The TIKI-100 system ROMs are bundled with the emulator and are **not copyrighted in a way that restricts redistribution** for this project. Original software (games, applications, etc.) on disk images may have its own copyright status — please respect the wishes of the original authors when downloading or sharing disks.

3. First start

3.1 Launching the emulator

The simplest possible launch is to start the emulator with a floppy disk already inserted in drive 0:

Windows (double-click): open the install folder and double-click `tiki100.exe`. The default boot disk is loaded automatically.

Windows (command prompt):

```
cd path\to\tiki100
tiki100.exe -fd0 disks\boot\tiko_kjerne_v4.01.dsk
```

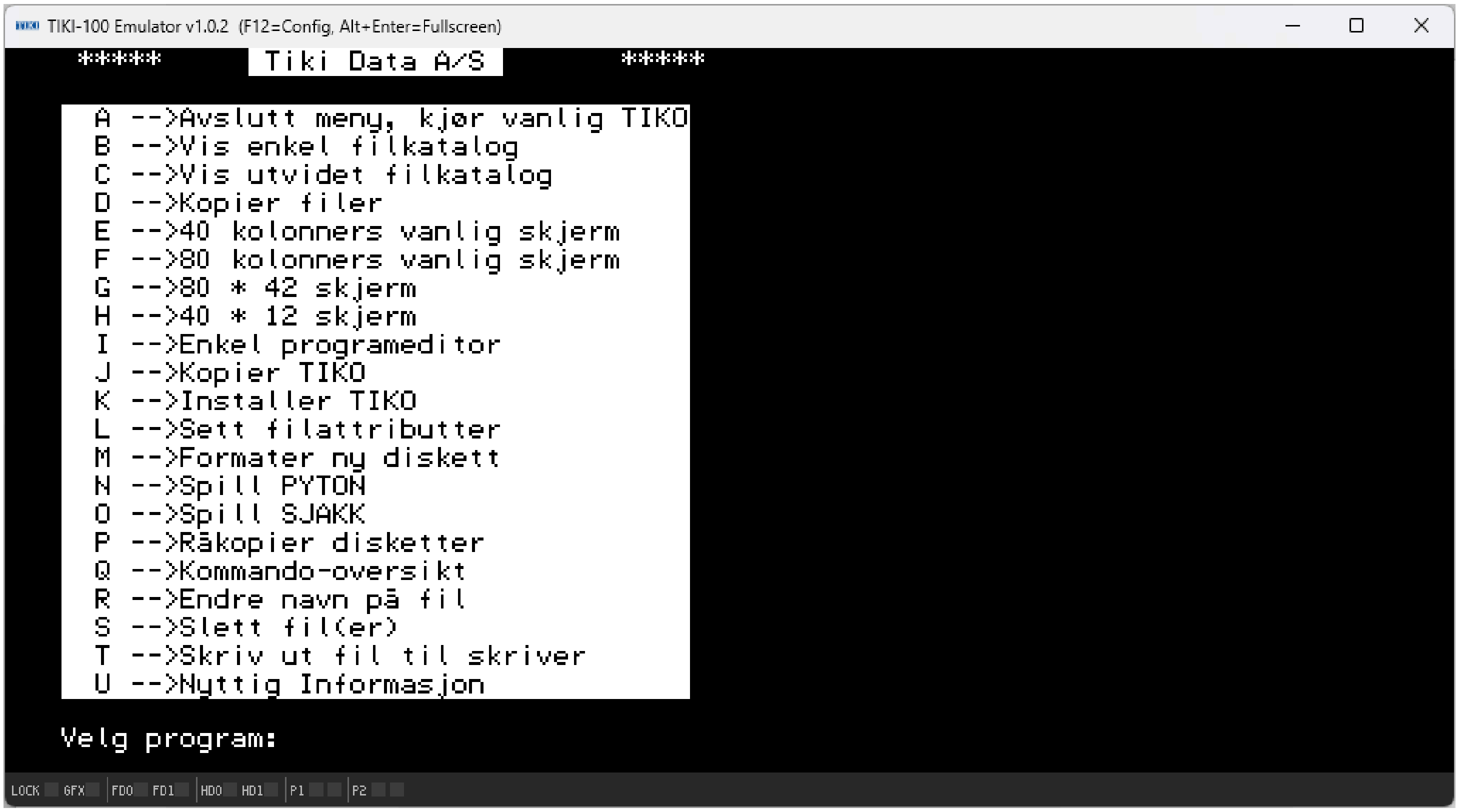
Linux / Raspberry Pi:

```
tiki100 -fd0 /usr/share/tiki100/disks/boot/tiko_kjerne_v4.01.dsk
```

macOS:

```
./tiki100 -fd0 disks/boot/tiko_kjerne_v4.01.dsk
```

After a moment, the emulator window appears, and the TIKI-100 boot screen shows up.



3.2 Understanding the main window

The window is divided into two regions:

```
+-----------------------------------------+
|                                         |
|    Emulated TIKI-100 screen             |
|    (text or graphics)                   |
|                                         |
+-----------------------------------------+
|  LOCK  GFX  FD0  FD1  HD0  HD1  | P1 P2 |   <- status bar
+-----------------------------------------+
```

- **Top area**: the TIKI-100's video output. Text mode characters, game graphics, BASIC prompts — all rendered here.
- **Bottom strip**: a status bar with small LEDs that mirror what the real machine would show. Hover the mouse over any LED for a tooltip in the window's title bar.

The LEDs are explained in detail in section 7.5.

3.3 Controls at a glance

Action	Key / control
Type into the TIKI-100	Just use your keyboard
Open the emulator's configuration menu	**F12**
Close the menu	**F12** or **Esc** (caution — Esc is also a TIKI-100 key)
Toggle fullscreen	**Alt + Enter**
Quit the emulator	Close the window

The mouse is only used inside the F12 configuration menu. The TIKI-100 itself was a keyboard-driven machine and the emulator does not pass mouse events to it.

4. Quick start guide

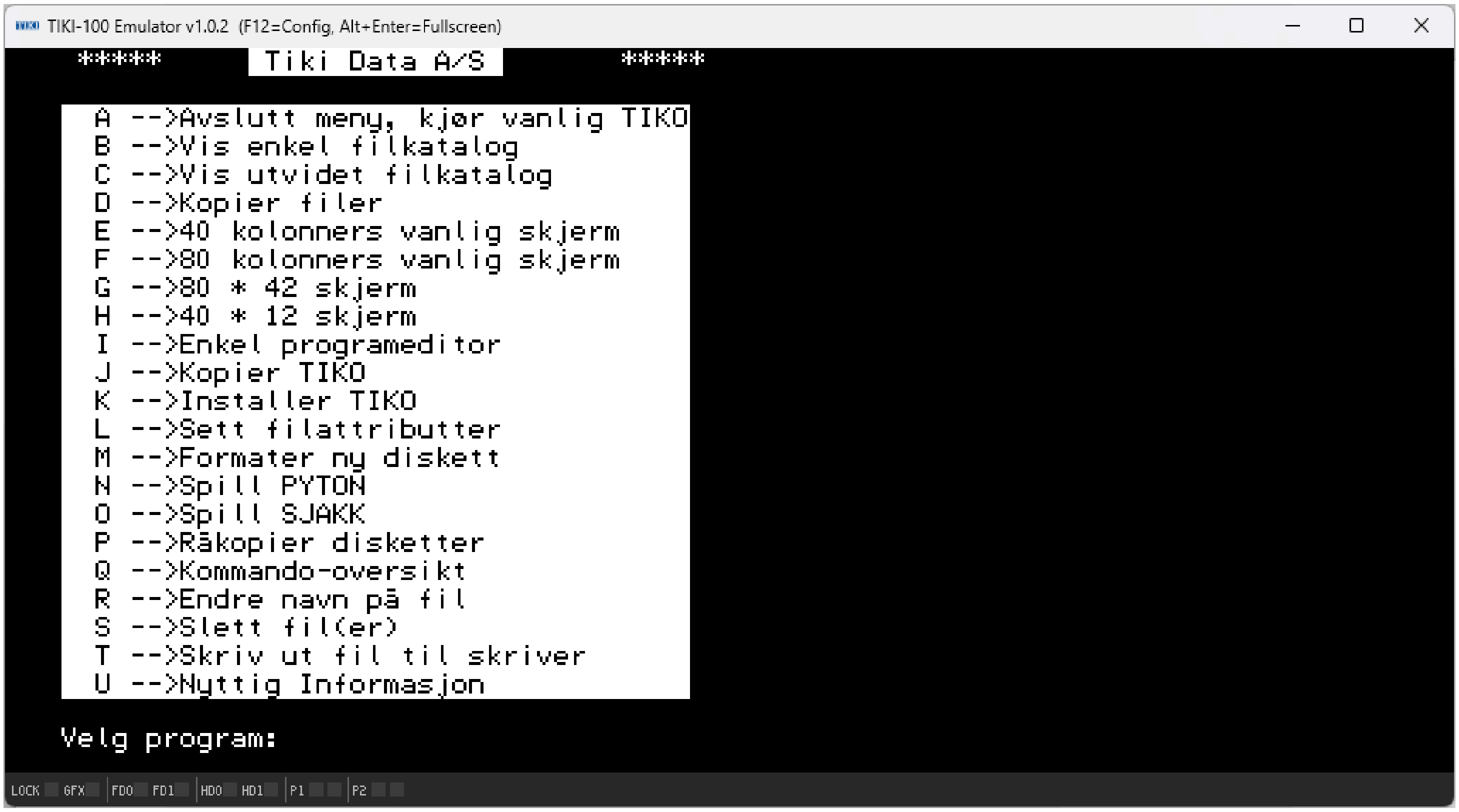
For users who want to be running software in five minutes.

Step 1 — Boot the emulator

Run the emulator with the bundled boot floppy:

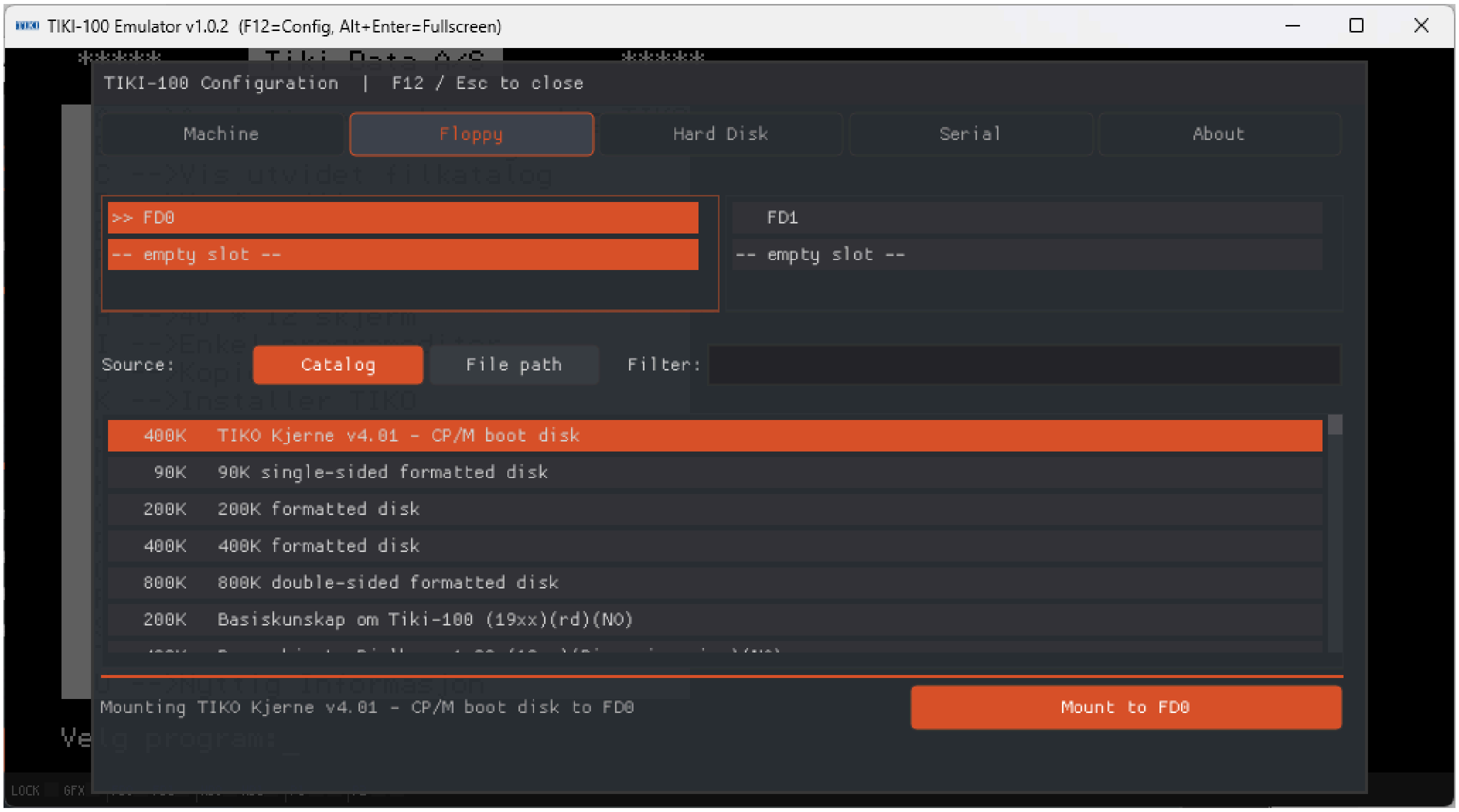
```
tiki100 -fd0 disks/boot/tiko_kjerne_v4.01.dsk
```

You will see the TIKI-100 sign-on, then the operating system prompt.



Step 2 — Mount a different disk

1. Press **F12**. The configuration overlay appears.
2. Click the **Floppy** tab at the top.
3. The list in the middle is the **disk catalogue** — every disk image the emulator knows about. Type in the **Filter** box to narrow the list, e.g. type `bas` to find BASIC-related disks.
4. Click a row to select it.
5. Click **Mount to FD0** (or **Swap into FD0** if a disk was already mounted) at the bottom right.



Step 3 — Use the disk

1. Press **F12** to close the menu.
2. Inside the TIKI-100, type the appropriate command for whatever's on the disk — for many CP/M disks, that's the program name (e.g. `WS` for WordStar, `BRUM`, etc.).
3. Press **Enter**.

If the disk is bootable, you may need to reboot the machine: press **F12** → **Machine tab** → **Reboot TIKI-100**.

Step 4 — Save your work

Saving works the way it does on the real TIKI-100: through the operating system or application running inside the emulator (e.g. `SAVE "FILENAME"` in BASIC). The data is written to the **disk image file** on your host machine — the `.dsk` file you mounted. Close the emulator and the file is automatically updated with any changes.

> **Tip — keep originals.** Make a copy of any `.dsk` you care about before mounting it, so you can roll back if something goes wrong.

5. Understanding disk images

5.1 What is a disk image?

A **disk image** is a single file that contains a byte-for-byte copy of a real floppy disk or hard disk. The TIKI-100 emulator reads from and writes to these files exactly as the real machine would read from real disks. The most common file extension is `.dsk`.

Two kinds of storage are emulated:

Type	Hardware emulated	Image size	File pattern
Floppy	FD1771 controller, 2 drives	90 KB to 800 KB per disk	`.dsk`
Hard disk	WD1010 controller, 2 drives	8 MB per drive	`.dsk` (typically `HD0.dsk`, `HD1.dsk`)

5.2 Drive naming

The emulator exposes drives by their **physical slot**:

- **FD0**, **FD1** — the two floppy drives.
- **HD0**, **HD1** — the two hard disk drives.

The TIKI-100 operating system then assigns logical drive letters (`A:`, `B:`, etc.) to whichever drives are present. So mounting an image to "FD0" is the physical operation; what drive letter the OS gives it depends on the OS configuration.

5.3 Mounting and ejecting disks

There are two ways to put a disk into a drive.

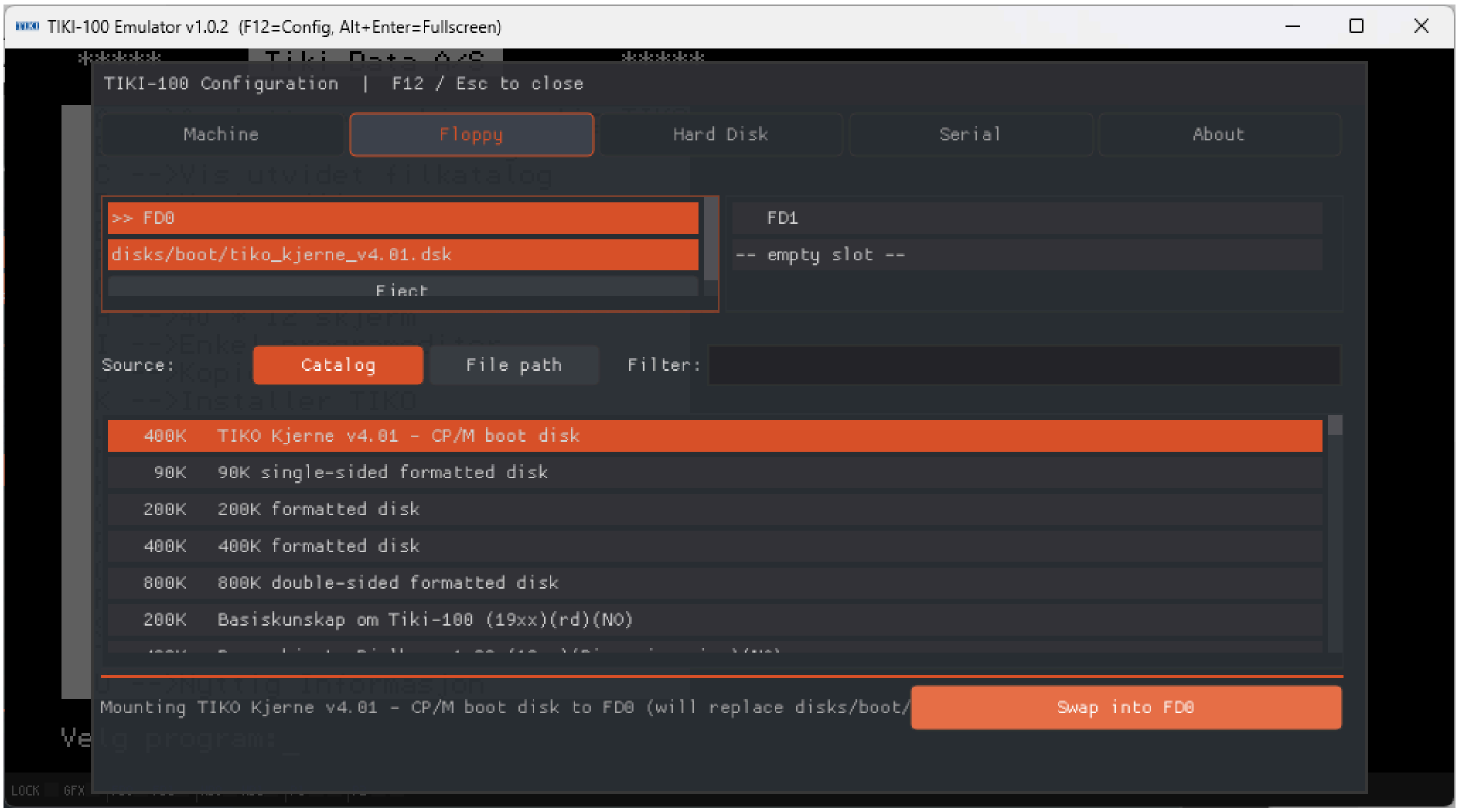
At launch, on the command line:

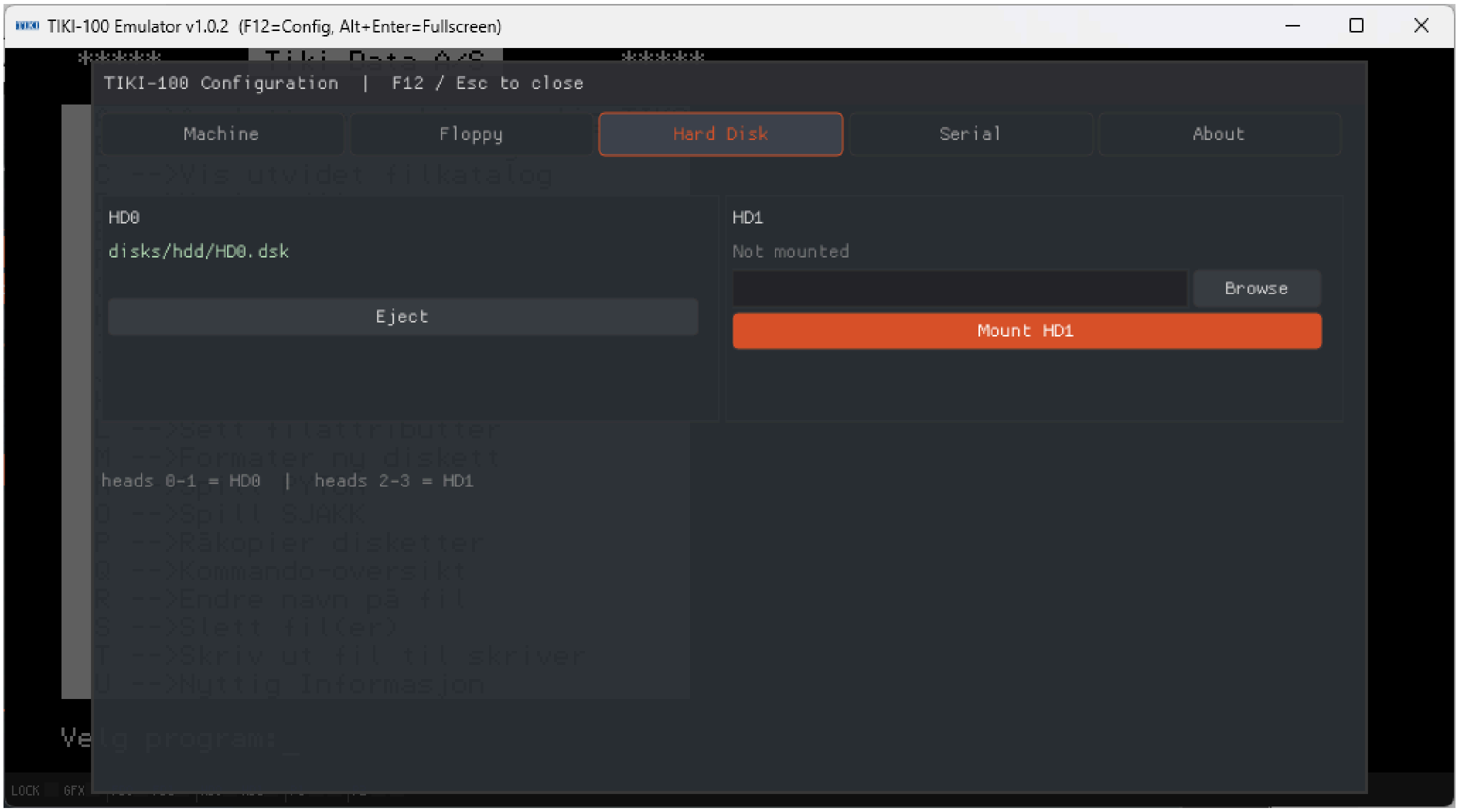
```
tiki100 -fd0 mygame.dsk -fd1 utils.dsk -hd0 HD0.dsk -hd1 HD1.dsk
```

At runtime, through the F12 menu:

1. Press **F12**.
2. Click the **Floppy** or **Hard Disk** tab.
3. Either pick from the **Catalog** (a curated list bundled with the emulator) or switch to **File path** and browse to a `.dsk` file anywhere on your computer.
4. Click **Mount**.

To **eject**, open the menu, find the drive card, and click **Eject**.





5.4 Where to find disk images

- The `disks/` folder bundled with the emulator contains a curated catalogue of period software.
- Online TIKI-100 archives (search for "TIKI-100 disk archive").
- Archives of CP/M software work too, since the TIKI-100 ran CP/M programs — but you may need a CP/M-compatible boot disk in FD0 first.

6. Keyboard and input

6.1 Direct keys

Most letter, number, and punctuation keys map directly. Just type and the characters appear, exactly as they would on the real TIKI-100.

6.2 Special TIKI-100 keys

The TIKI-100 keyboard has a few keys with no obvious modern equivalent:

Press on your PC	TIKI-100 key	What it does
Esc	ESC (which on the real machine was **CTRL+Æ**)	Sends the ESC code
F10 or **Pause**	BRYT	Break key — interrupts the running program
Backspace or **Delete**	SLETT	Delete character
F8	GRAFIKK	Toggles graphics mode

> **Why isn't Esc on the real keyboard?** The original TIKI-100 keyboard simply did not have an ESC key — instead, the manual documented **CTRL + Æ** as the ESC sequence. The emulator translates your modern Esc key into that combination automatically.

6.3 Emulator-level shortcuts

These shortcuts are handled by the emulator itself and are **not** sent to the TIKI-100:

Shortcut	What it does
F12	Open the configuration overlay (see section 7.1)
Esc (when overlay is open)	Close the overlay
Alt + Enter	Toggle fullscreen

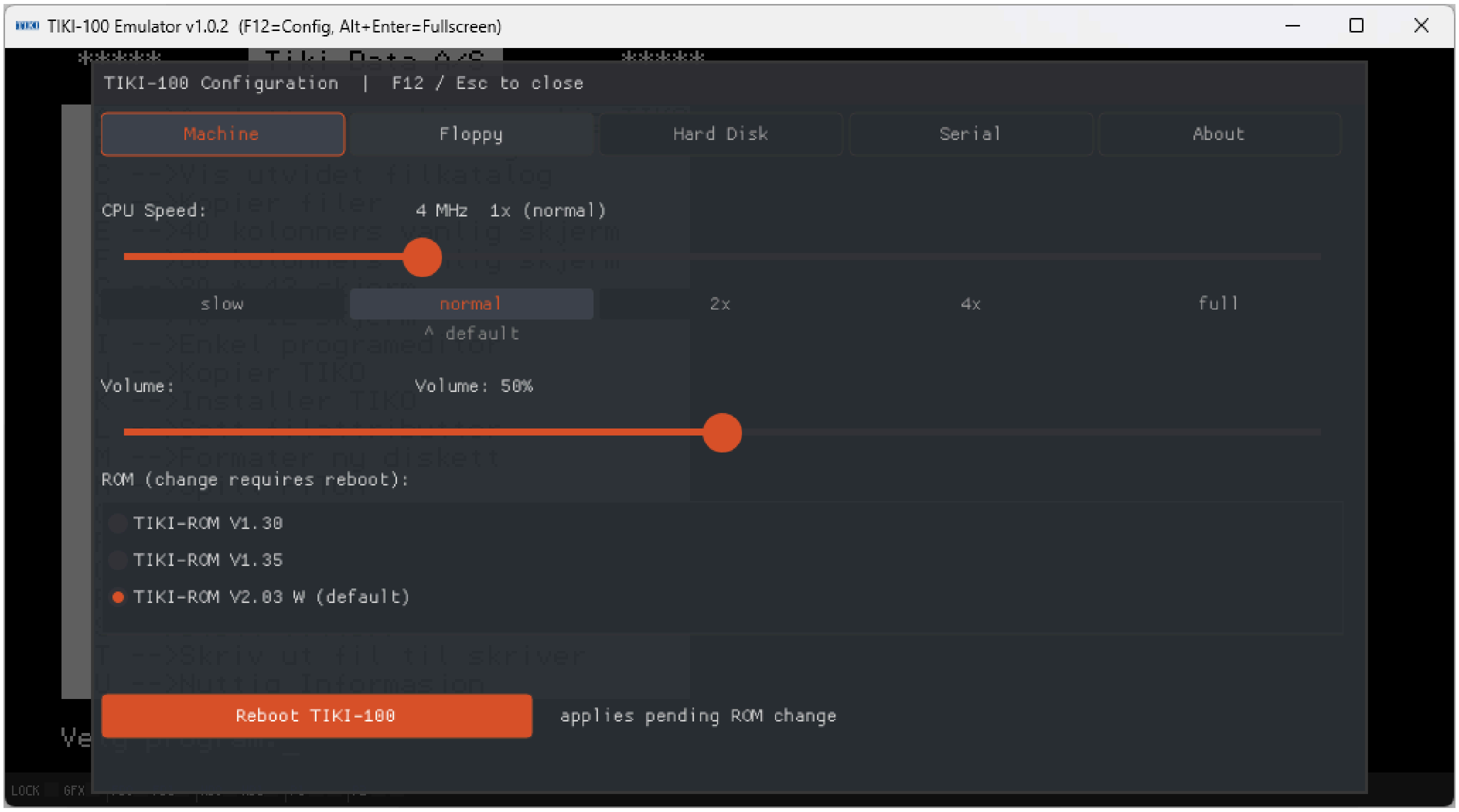
7. Emulator features

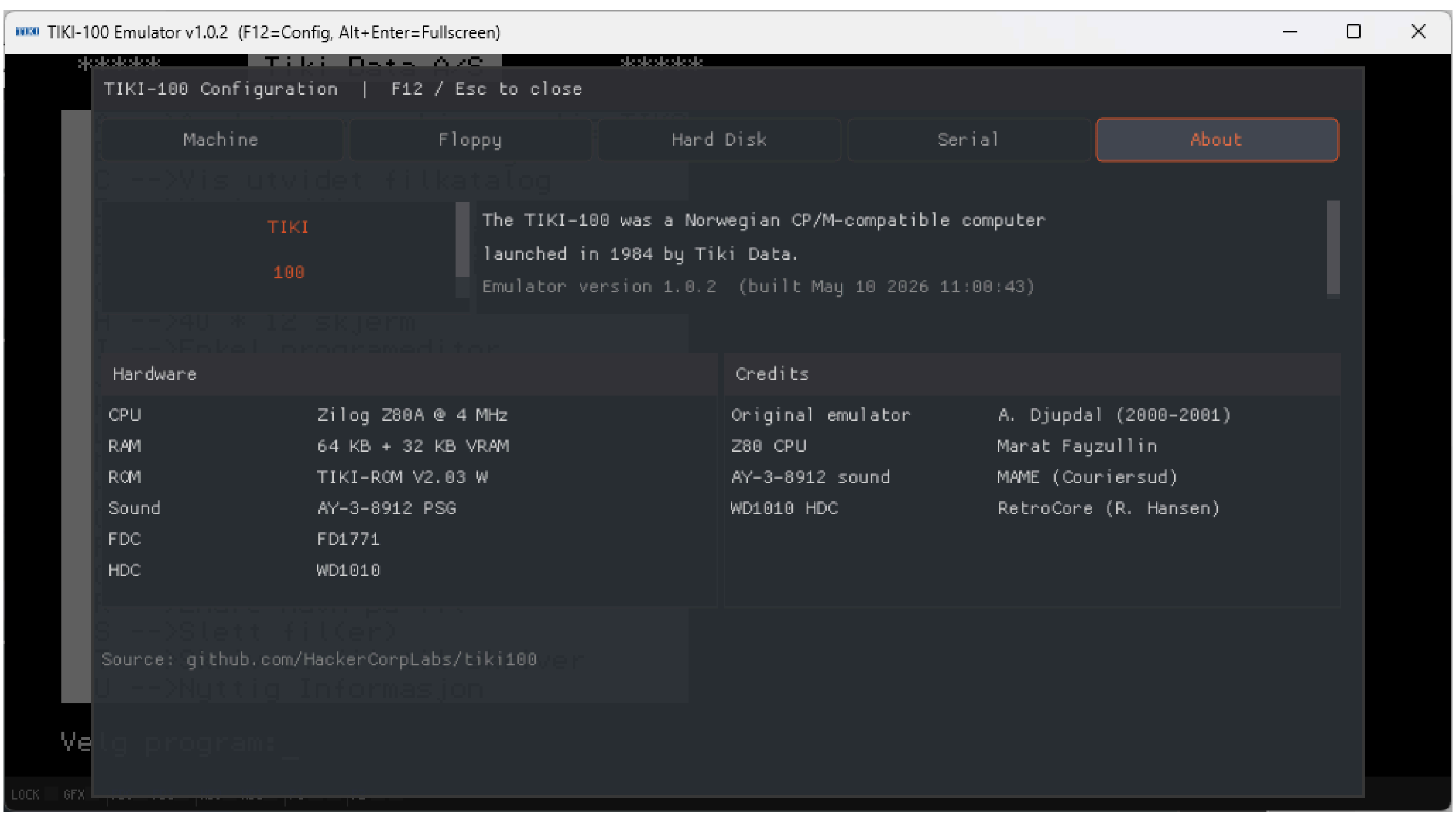
7.1 Configuration overlay (F12 menu)

Press **F12** at any time to pause the emulation and open the on-screen configuration overlay. It has five tabs:

- **Machine** — CPU speed, volume, ROM selection, soft reset.
- **Floppy** — Mount/eject floppy disks; pick from a catalog or browse.

- **Hard Disk** — Mount/eject 8 MB hard disk images.
- **Serial** — Configure the two emulated serial ports.
- **About** — Version and credits.





Every control on every tab is documented in detail in MENU.md. The most common things you'll touch are covered below.

7.2 Speed control

In the **Machine** tab the CPU speed slider has five stops:

Setting	Equivalent CPU clock	Use when
`slow`	2 MHz	You want to slow things down to read scrolling text
`normal`	4 MHz (default — period-accurate)	Authentic experience
`2x`	8 MHz	Speeds up long compiles or disk operations
`4x`	16 MHz	Very fast — useful for tedious tasks
`full`	unthrottled	Runs as fast as your modern CPU allows; sound may stutter

You can also change speeds by clicking any of the labels under the slider (`slow` , `normal` , `2x` , `4x` , `full`). The active setting is shown in orange.

7.3 Volume

Below the speed slider is a **Volume** slider (0% – 100%). It controls the AY-3-8912 sound chip output. Drag the knob, click anywhere on the track, or just leave it at the default 50%.

7.4 Reset (soft reboot)

In the **Machine** tab there is a **Reboot TIKI-100** button. This performs a soft reset of the Z80 — equivalent to pressing the reset switch on the real machine. It is needed:

- After changing the active ROM.
- If the running OS hangs.
- To boot from a different floppy without quitting the emulator.

> **Note:** Reset does **not** unmount disks. Whatever is mounted in FD0/FD1/HD0/HD1 stays put across the reboot.

7.5 Status bar LEDs in detail

The status bar at the bottom of the window mirrors the real TIKI-100's front-panel indicators, plus extras for the emulated serial ports.

System LEDs (left group):

LED	When lit
LOCK (green)	Caps Lock is active
GFX (cyan)	Graphics mode is enabled (video RAM mapped into Z80 address space)
FD0 (yellow)	Floppy drive 0 is currently selected and active
FD1 (yellow)	Floppy drive 1 is currently selected and active
HD0 (green)	Hard disk 0 is being accessed (stays lit briefly after each access)
HD1 (green)	Hard disk 1 is being accessed (stays lit briefly after each access)

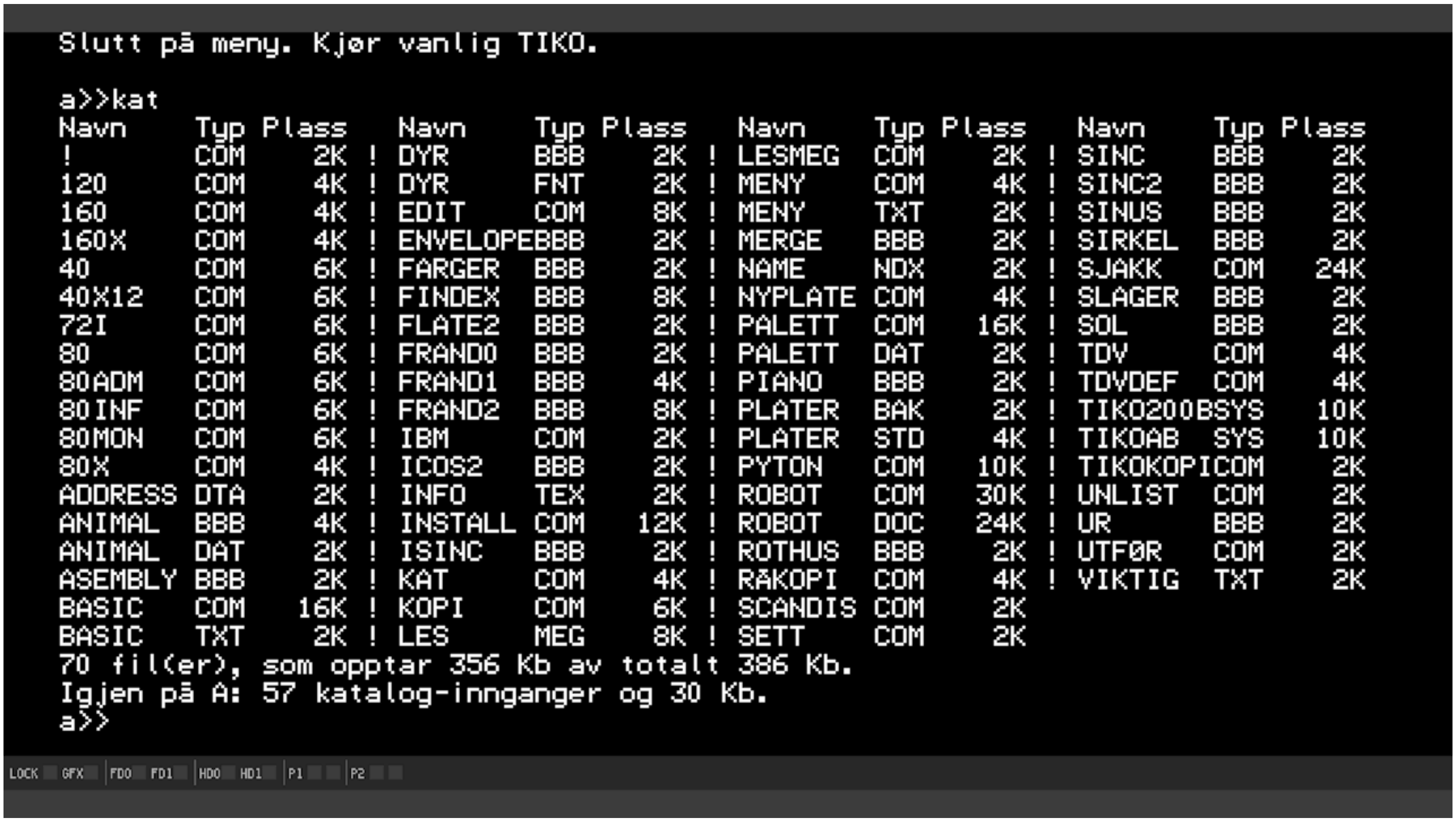
Serial LEDs (right of the `|` separator):

LED	Meaning
P1 TX (orange)	Serial channel A transmitted a byte
P1 RX (cyan)	Serial channel A received a byte
P2 TX (orange)	Serial channel B transmitted a byte
P2 RX (cyan)	Serial channel B received a byte

LEDs flash for ~150 ms after each byte. Hover your mouse over a LED to see a tooltip in the title bar.

7.6 Fullscreen

Press **Alt + Enter** to toggle borderless fullscreen mode. Press it again to return to a window.



```
Slutt på meny. Kjør vanlig TIKO.

a>>kat
Navn    Typ Plass   Navn    Typ Plass   Navn    Typ Plass   Navn    Typ Plass
!       COM    2K ! DYR     BBB    2K ! LESMEG  COM    2K ! SINC    BBB    2K
120     COM    4K ! DYR     FNT    2K ! MENY    COM    4K ! SINC2   BBB    2K
160     COM    4K ! EDIT    COM    8K ! MENY    TXT    2K ! SINUS   BBB    2K
160X    COM    4K ! ENVELOPEBBB    2K ! MERGE   BBB    2K ! SIRKEL  BBB    2K
40      COM    6K ! FARGER  BBB    2K ! NAME    NDX    2K ! SJAKK   COM   24K
40X12   COM    6K ! FINDEX  BBB    8K ! NYPLATE COM    4K ! SLAGER  BBB    2K
72I     COM    6K ! FLATE2  BBB    2K ! PALETT  COM   16K ! SOL     BBB    2K
80      COM    6K ! FRAND0  BBB    2K ! PALETT  DAT    2K ! TDV     COM    4K
80ADM   COM    6K ! FRAND1  BBB    4K ! PIANO   BBB    2K ! TDVDEF  COM    4K
80INF   COM    6K ! FRAND2  BBB    8K ! PLATER  BAK    2K ! TIKO200BSYS   10K
80MON   COM    6K ! IBM     COM    2K ! PLATER  STD    4K ! TIKOAB  SYS   10K
80X     COM    4K ! ICOS2   BBB    2K ! PYTON   COM   10K ! TIKOKOPICOM    2K
ADDRESS DTA    2K ! INFO    TEX    2K ! ROBOT   COM   30K ! UNLIST  COM    2K
ANIMAL  BBB    4K ! INSTALL COM   12K ! ROBOT   DOC   24K ! UR      BBB    2K
ANIMAL  DAT    2K ! ISINC   BBB    2K ! ROTHUS  BBB    2K ! UTF0R   COM    2K
ASEMBLY BBB    2K ! KAT     COM    4K ! RAKOPI  COM    4K ! VIKTIG  TXT    2K
BASIC   COM   16K ! KOPI    COM    6K ! SCANDIS COM    2K
BASIC   TXT    2K ! LES     MEG    8K ! SETT    COM    2K
70 fil(er), som opptar 356 Kb av totalt 386 Kb.
Igjen på A: 57 katalog-innganger og 30 Kb.
a>>
```


7.7 Audio

Audio comes out of your default system audio device automatically. There is no input/device selector inside the emulator — if you need to route audio to a specific device, use your operating system's normal audio routing (System Settings on macOS, Sound on Windows, PulseAudio / PipeWire on Linux).

7.8 Persistence

Persistence is handled the way it was on the real machine — by saving files to a mounted disk image. The disk image is written back to the host filesystem when you save inside the running OS, so your work survives quitting and re-launching.

8. Common software

The TIKI-100 ran a wide mix of native Norwegian software and international CP/M titles. The list below is a starting point, not exhaustive.

8.1 Operating systems

- **TIKO/KP/M** — The native operating system. CP/M-compatible. This is what boots from `tiko_kjerne_v4.01.dsk`.

8.2 Programming languages

- **TIKI-BAS** — Native BASIC interpreter, similar in feel to other early-1980s BASICs.
- **Turbo Pascal 3.0** — A classic Pascal compiler that was hugely popular under CP/M.
- **Microsoft BASIC, FORTRAN, COBOL** — All available in CP/M form.
- **Z80 assembler** (M80/L80, ASM/HEX, etc.) — Period-authentic assembly toolchains.

8.3 Productivity

- **WordStar** — The dominant CP/M word processor of the era.
- **SuperCalc** — Spreadsheet (the CP/M equivalent of VisiCalc).
- **dBase II** — Database.
- **Tiki-Kalk** — Native TIKI-100 spreadsheet.

8.4 Educational and Norwegian-specific

- **BRUM** — A native TIKI-100 learning environment used in Norwegian schools.
- Numerous teacher-authored programs from the Norwegian school project.

8.5 Games

A range of CP/M games (text adventures, simple action games, puzzles) will run. The TIKI-100 was primarily a serious productivity machine, so the games library is smaller than for, say, the ZX Spectrum.

9. Troubleshooting

9.1 The emulator will not start

Windows: "SDL2.dll was not found"

The `tiki100.exe` was moved away from `SDL2.dll`. Either move it back, or copy `SDL2.dll` next to wherever the EXE now lives.

macOS: "tiki100 cannot be opened because the developer cannot be verified"

The binary is unsigned. Either right-click → **Open** in Finder once, or run `xattr -d com.apple.quarantine tiki100` in Terminal.

macOS: "dyld: Library not loaded"

SDL2 is not installed. Run `brew install sdl2`.

Linux: "command not found"

After a `.deb` install, `tiki100` should be on your PATH automatically. Try opening a fresh terminal, or call it explicitly: `/usr/bin/tiki100`.

9.2 Black or empty screen

- **No `-fd0` flag:** the emulator started without a boot disk. Quit and re-launch with a floppy image (see section 4).
- **Headless Raspberry Pi**: install a graphical desktop or boot into one. SSH alone is not enough.
- **Stuck during boot**: try the Reboot button in F12 → Machine tab.

9.3 Keyboard problems

- **F-keys do nothing** — make sure your operating system isn't intercepting them (e.g. the Mac touch bar or a laptop's "Fn" lock).
- **Esc closes the menu instead of being sent to the TIKI-100** — yes, this is intentional when the F12 menu is open. Close the menu first, then press Esc.

9.4 Disk errors

- **"Disk full" inside the OS** — the emulated disk image really is full. Delete files inside the OS, or mount a fresh blank image.
- **Changes are lost after quitting** — make sure you saved inside the running program before quitting. The emulator does not auto-save RAM contents.
- **"File not found"** when mounting — double-check the path and that the file actually exists. Use the **Browse** button rather than typing the path manually.

9.5 Performance issues

- **Sound stutters or game runs too slow** — your host CPU may be struggling. On a Raspberry Pi Zero 2 / Pi 2, try the `-fast` flag, or set the Speed slider to **full**.
- **Game runs too fast** — set Speed to **normal** (the default). Modern CPUs are far faster than 4 MHz Z80s, so without throttling the emulator runs much faster than the real machine.

9.6 The F12 menu is small in the corner when the window is maximized

Known cosmetic issue. The menu stays at its native size in the top-left while the rest of the window is dimmed. All buttons remain clickable; they're just not visually scaled up. See MENU.md — Known issues.

10. Advanced usage

10.1 Command-line flags

The full reference is in USAGE.md. Quick summary:

```
tiki100 [options]

  -fd0 <file>      Mount floppy image in FD0
  -fd1 <file>      Mount floppy image in FD1
  -hd0 <file>      Mount hard disk image in HD0
  -hd1 <file>      Mount hard disk image in HD1
  -scale <1-4>     Initial window scale (1 = native, 2 = 2x, etc.)
  -fast            Run at full speed (no throttle, sound may drift)
  -sera <mode>     Configure serial channel A (modem | listen:PORT | connect:HOST:PORT)
  -serb <mode>     Configure serial channel B
  -h, --help       Show help
```

Examples:

```
# Boot floppy + 2 hard disks
tiki100 -fd0 disks/boot/tiko_kjerne_v4.01.dsk -hd0 HD0.dsk -hd1 HD1.dsk

# Larger window
tiki100 -scale 2 -fd0 boot.dsk

# Channel A listens on TCP 5000 for incoming connections
tiki100 -sera listen:5000 -fd0 boot.dsk
```

10.2 Custom ROMs

The emulator ships with three ROMs:

File	Label
`tikirom-1.30`	TIKI-ROM V1.30
`tikirom-1.35`	TIKI-ROM V1.35
`tikirom-2.03w`	TIKI-ROM V2.03 W *(default)*

To add another ROM:

1. Drop the binary file into the `rom/` folder (8 KB, raw binary).
2. Edit `rom/roms.json` and add an entry, e.g.:

   ```
   {"file": "my-custom-rom", "label": "My Custom ROM"}
   ```

3. Restart the emulator.
4. Open F12 → Machine tab and pick your new ROM in the list.
5. Click **Reboot TIKI-100** to load it.

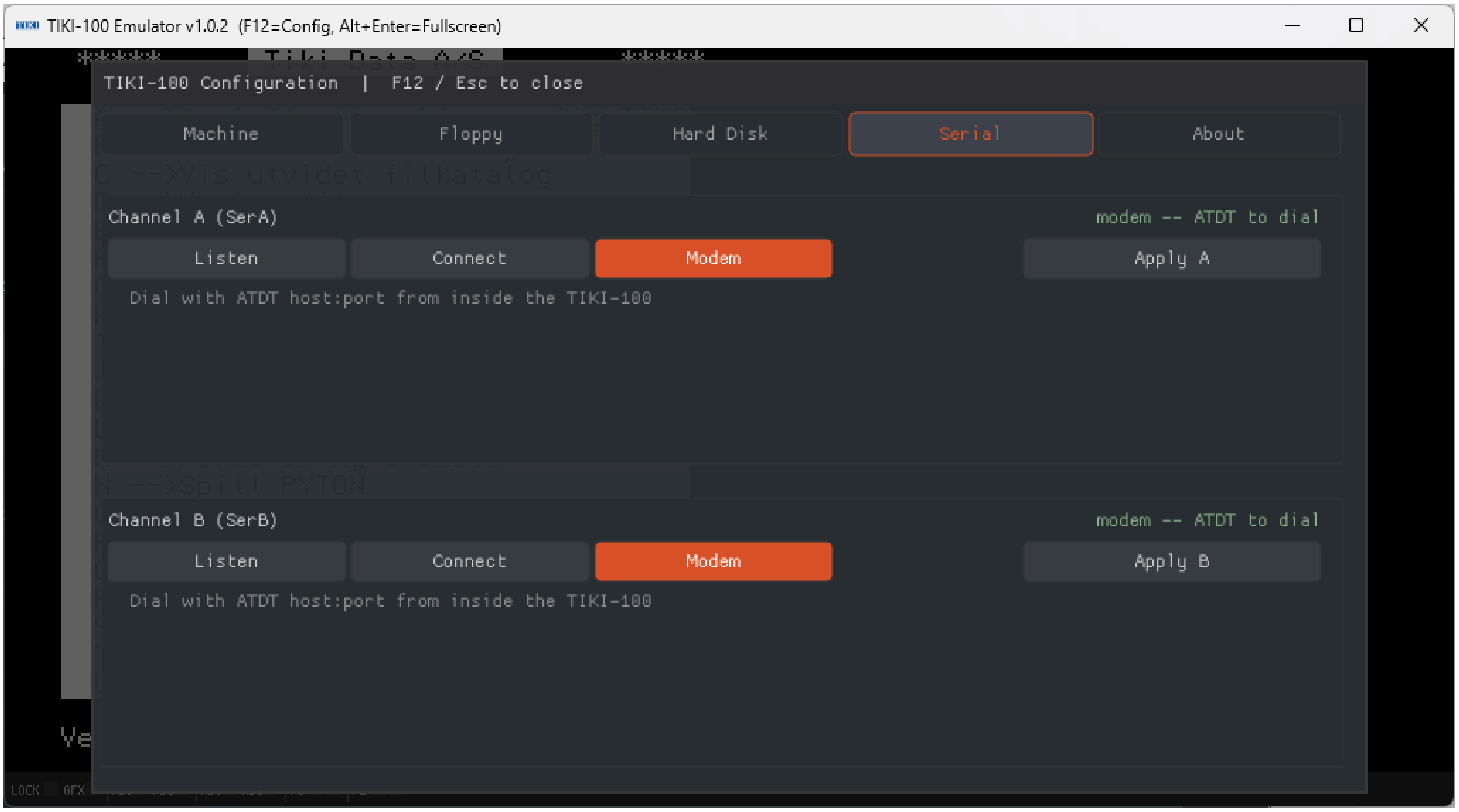
10.3 Custom disk catalog

The Floppy tab's Catalog is read from `disks/disks.json`. You can edit this file to add your own disks to the menu. The format is self-documenting — open the existing file in a text editor.

10.4 Serial ports — TCP networking

Each serial port can be set to one of three modes (in the F12 menu's Serial tab, or with `-sera` / `-serb` on the command line):

- **Modem** *(default)* — Hayes-compatible. Dial out from a TIKI-100 terminal program with `ATDT host:port`. Lets you connect to BBSes, echo servers, or other TIKI-100s over the internet.
- **Listen** — TCP server. Bind a local port; the next incoming connection is forwarded straight to the DART chip. Useful for hosting your own service.
- **Connect** — TCP client. Permanently connected to a fixed host:port, with auto-reconnect. Useful for a dedicated link.



The full reference (including the entire AT command set) is in SERIAL.md.

10.5 Hardware emulation details

For developers, hardware hackers, or the curious, the I/O port map and chip-level behaviour are documented in HARDWARE.md.
Highlights:

- Z80A CPU at 4 MHz
- 64 KB main RAM + 32 KB graphics RAM
- 8 KB monitor ROM at `0xF000-0xFFFF` (when ROM-enabled)
- Three video modes (1024×256/2-col, 512×256/4-col, 256×256/16-col)
- AY-3-8912 PSG (3 tone + noise + envelope)
- FD1771 floppy controller (2 drives)
- WD1010 hard disk controller (2 × 8 MB drives)
- Z80 DART (2 serial channels), Z80 PIO (parallel/printer), Z80 CTC (4-channel timer)

11. FAQ

Q: Is the emulator free?
Yes. The source is open and the pre-built binaries are free downloads. See CREDITS.md for the per-component licenses.

Q: Does it run real TIKI-100 software?
Yes — that is the entire point. Original `.dsk` images boot and run unmodified.

Q: Will my saved work survive quitting?
Yes, provided you saved inside the running OS to a mounted disk. The disk image file on your host is updated as you save.

Q: Can I run CP/M programs that didn't ship for TIKI-100?
Yes, in many cases. Boot a CP/M-compatible disk in FD0, then mount the program's disk in FD1 and access it with the appropriate drive letter.

Q: Is there a Mac Intel build?
The published macOS build is **arm64 only** (Apple Silicon).

Q: How do I uninstall?

- **Windows:** delete the extracted folder.
- **Linux/Pi:** `sudo apt remove tiki100` .
- **macOS:** delete the extracted folder and (optionally) `brew uninstall sdl2` .

12. Glossary

Z80 / Z80A — The 8-bit CPU at the heart of the TIKI-100, made by Zilog and released in 1976. Hugely influential in 1980s computing.

CP/M — *Control Program for Microcomputers.* The dominant operating system for 8-bit business computers before MS-DOS took over. The TIKI-100's TIKO/KP/M is a compatible variant.

ROM — *Read-Only Memory.* A chip containing the system firmware (monitor, boot routines). Can't be changed by the running computer.

RAM — *Random Access Memory.* Working memory, lost when the power goes off (or when you quit the emulator without saving).

Floppy disk image / `.dsk` file — A single file on your modern computer that contains all the bytes that would have been on a real floppy disk. The emulator treats it exactly as if it were a floppy inserted in a drive.

Mount / Unmount (Eject) — "Mount" means tell the emulator that a particular disk image is now in a particular drive slot. "Unmount" or "Eject" reverses this.

FD0 / FD1 — The two **physical floppy drives** the emulator exposes. The TIKI-100's OS may give them logical drive letters internally.

HD0 / HD1 — The two **physical hard disk drives**. Each can hold an 8 MB image.

Emulator — A program that pretends to be different hardware. The TIKI-100 emulator pretends to be a 1984 TIKI-100 on your modern PC.

SDL2 — A library used by the emulator for the window, keyboard, sound, and timing. On Windows it ships as `SDL2.dll` next to the EXE; on Linux/macOS you install it separately.

Hayes / AT commands — A standard command set for controlling modems, named after the company Hayes Microcomputer Products. The emulator's serial channels can pretend to be a Hayes modem so software that knows how to dial a real modem can dial out over TCP.

BBS — *Bulletin Board System.* Pre-internet text-based dial-in services. Some BBSes are still online today and can be dialled with the emulator's modem mode.

DART / PIO / CTC — Zilog support chips. DART = dual serial; PIO = parallel I/O; CTC = counter/timer. All three are emulated.

FD1771 / WD1010 — The Western Digital floppy and hard disk controller chips that the TIKI-100 used. Both are emulated.

AY-3-8912 — The General Instrument programmable sound generator chip. Three tone channels, one noise channel, an envelope generator. Used in many 1980s machines.

Speed throttle — Software that slows down the emulator so that one emulated second equals one real second. With throttle off (`-fast` or the **full** speed setting) the emulator runs as fast as your CPU allows.

13. Additional resources

Project documentation

- **INSTALL.md** — Installing pre-built binaries.
- **USAGE.md** — Command-line flags, key reference, status LED reference.
- **MENU.md** — Every control in the F12 menu, in detail.
- **SERIAL.md** — TCP serial modes and the Hayes AT command set.
- **HARDWARE.md** — Hardware emulated, I/O port map, source-tree layout.
- **CREDITS.md** — Acknowledgements and licenses.
- **apple-dev.md** — Notes on signing and notarizing the macOS build.

About the original machine

- **Norsk Teknisk Museum** — historical pages on the TIKI-100 (in Norwegian).
- The Wikipedia entry for **TIKI-100** has a good summary in English.
- Period magazine archives (search for "Tiki Data" + the magazines of the era).

Project on GitHub

- Issues, pull requests, and the full source code: `https://github.com/HackerCorpLabs/tiki100`

- Releases (pre-built binaries):
  `https://github.com/HackerCorpLabs/tiki100/releases`

Acknowledgements

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Full path of this document:

`E:\Dev\Emulators\Z80\ronny-tiki100\docs\MANUAL.md`